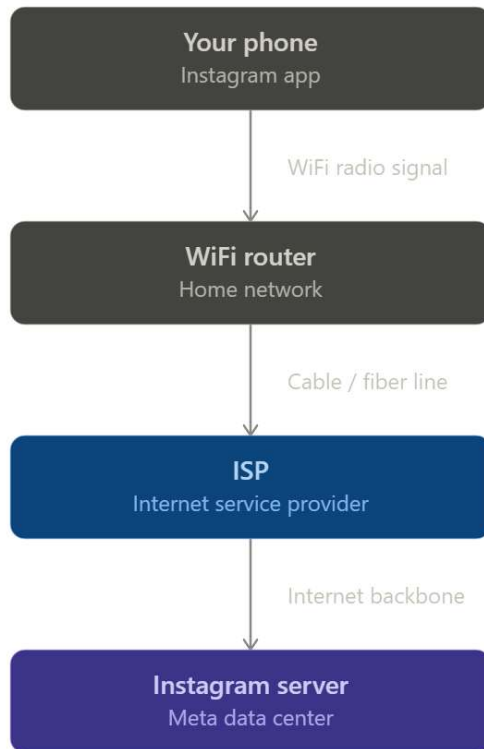


INTRODUCTION TO NETWORKING

Task 1 : Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



- **Phone → WiFi router:** your phone sends the request over a radio signal (2.4 GHz or 5 GHz).
- **Router → ISP:** the router forwards it out through a cable or fiber line to your internet service provider's network.
- **ISP → Instagram server:** the ISP routes it across the internet backbone to whichever Meta data center is serving Instagram, which sends the response (your feed, photos, etc.) all the way back the same path.

The **internet backbone** is the network of high-capacity fiber optic cables and routers that connect major networks (ISPs, data centers, countries) together across long distances.

Task 2: List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

LAN – Local area networks

WAN – Wide area networks

PAN – Personal Area Network

1. Smartphone:

Connects to: WLAN – WiFi at home, part of a LAN and WAN – mobile data network like 4G/5G. It also uses PAN when paired with Bluetooth earphones or a smartwatch.

2. WiFi Router:

Connects to: LAN - creates the local network for all home devices and WAN – connects to the ISP to bring internet into the home. This makes it the bridge device between your LAN and the WAN.

3. Laptop

Connects to: LAN via WiFi or Ethernet cable to the home/office network which then reaches the WAN (internet) through the router.

4. Smartwatch

Connects to: PAN (Personal Area Network) — it typically pairs with your smartphone over Bluetooth, which is a short-range network just for your personal devices. Some models also connect directly to WLAN/WAN if they have built-in WiFi or SIM support.

5. Smart TV

Connects to: LAN – connects to your home WiFi network to reach the WAN (internet) for streaming services like YouTube or Netflix.

Task 3: Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Internet — the global, public network of networks (World Wide Web, email, cloud services) accessible to anyone with connectivity.

Intranet — a private network restricted to an organization (employees or authorized members) used for internal communication, resources and apps.

Example: College Network

Think about your IT course setup:

Intranet

example:

Tops Technologies has internal portal — 'Career Center', where you check attendance, download study materials, view exam results, or submit assignments. This is only accessible:

- While connected via internal LAN, or
- Through a login with your student ID/credentials from outside

This portal is **not** something a random person on the internet can just visit and use. It's built for the students of TOPs Technologies and staff — that's what makes it an intranet.

Internet

example:

When you're on that same college WiFi and you open Google, YouTube, you're going out through the college's router, past their ISP, onto the **public Internet** — accessible to anyone in the world, not just college members.

Task 4: Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app. Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

Example used:

APP – Zomato

ISP – Jio fiber

Stage 1: Opening the Zomato Application

1. When the Zomato application is launched on a device connected to a home WiFi network, the request first travels from the smartphone to the WiFi router (forming the LAN).

2. The router forwards this request through the Jio Fiber ONT (Optical Network Terminal), which connects the home network to Jio Fiber's network — acting as the Internet Service Provider (ISP). From there, the request travels onto the Internet backbone.

- **ONT:** Converts the optical (light) signal from the fiber cable into a digital electrical signal that networking equipment can process.
- **Router:** Takes that digital signal from the ONT and creates the LAN — assigning IP addresses to devices, managing traffic, and broadcasting the WiFi signal so phones/laptops can connect wirelessly.

3. A Domain Name System (DNS) server translates "zomato.com" into its IP address, after which the smartphone establishes a connection with Zomato's server.

- By default, when the device is connected to the internet, it automatically uses Jio's own DNS servers to resolve domain names like "zomato.com" into IP addresses. This happens automatically unless its manually changed in the DNS settings.

4. The server responds with home-screen data — restaurant listings, banners, and offers — which travels back along the same path (server → backbone → Jio Fiber network → ONT → router → smartphone) to be displayed on the screen.

Stage 2: Performing a Search

1. When the user enters a search term such as "biryani" and taps search, the smartphone sends the query along with GPS location data through the same path: smartphone → router (LAN) → Jio Fiber ONT → Jio Fiber's ISP network → Internet backbone → Zomato's server. The DNS lookup is not repeated at this stage, since it was already cached during the initial connection. Zomato's server queries its database for nearby matches and may also contact a maps/location API to calculate distance and delivery time. The results are transmitted back along the same path and displayed on the screen within one to two seconds.